Tejus Samir

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EDUCATION

Texas A&M University - College Station Expected Graduation: May 2029
Bachelor of Science in Electrical Engineering Classification: Junior | GPA: 3.7/4.0
Extracurriculars: Ignitors Rocketry Team, IEEE, Texas Student Rocket Propulsion Laboratory
Authorized to work in the U.S. without sponsorship

EXPERIENCE

Ignitors Rocketry Team, Flight Computer Team - College Station, TX *Dec 2025 – Present*

- Designed a pyro firing and continuity-sensing stage for a custom flight computer, using low-side N-channel MOSFET switching to deliver ~3 A per channel from 3.3 V microcontroller logic
- Implemented low-side N-channel MOSFET power switching circuits to deliver ~3 A of firing current per channel to e-match loads directly from 3.3 V MCU logic

Texas Student Rocket Propulsion Laboratory, Avionics Researcher - College Station, TX *Aug 2026 – Present*

- Designing a dual flight computer avionics system with a custom backup computer, doubling pyro channels and adding 6DOF sensing for air-start control and 500 Hz data logging
- Leading development on a custom Teensy 4.1–based backup flight computer integrated with GPS and an onboard camera, providing redundant recovery, location tracking, and flight video
- Modeling the avionics bay from scratch in Onshape (3D CAD) with the team, integrating both flight computers, power, and wiring within a low power, 2-battery budget and a size-constrained airframe

PROJECTS

FreshPal Smart Sensor | Altium Designer, ESP32, I2C, PCB Design, Embedded Systems *Aug 2023 – May 2025*

- Designed a custom 2-layer PCB in Altium Designer integrating an ESP32 microcontroller, an I2C temperature/humidity sensor, and a Winsen GM-802B MEMS ammonia (NH_3) sensor to detect food spoilage
- Developed the NH_3 sensing circuitry, including heater drive and voltage-divider signal conditioning, and programmed C++ firmware in Arduino IDE using non-blocking architecture, I2C communication, and Wi-Fi data transmission
- Authored a 22-page feasibility/technical plan earning 3rd place internationally at the DECA International Career Development Conference among 2,500+ competitors

Audio Signal Conditioning Amplifier | TINA-TI, Altium Designer *Aug 2026*

- Designed and simulated an inverting op-amp amplifier in TINA-TI with a gain of ~14.6 V/V (23.3 dB) and a ~1.06 kHz active low-pass anti-aliasing filter
- Laid out circuit as a PCB in Altium Designer, translating simulated design into a physical board

Counterfeit Semiconductor Detection | Arduino, MATLAB, Python, Semiconductor Validation *Apr 2026*

- Built an Arduino/ADS1115 test bench sweeping all 4,096 codes of a 12-bit DAC and a MATLAB statistical model that detects counterfeit chips by their DNL/INL nonlinearity
- Presented to a panel of industry judges to win 1st place and a $5,000 grand prize at Aggies Invent: Hardware Resilience

SKILLS AND CERTIFICATIONS

Awards/Honors: Eagle Scout (Scouting America), Aggies Invent - 1st Place, DECA Internationals - 3rd Place
Electronics Design & Simulation: Altium Designer, LTspice, Onshape, TINA-TI, OpenRocket, PCB Layout
Embedded Systems: ESP32, Teensy 4.1, Arduino, I2C, Embedded C/C++, ADS1115 ADC
Programming: Python, Go, MATLAB, C++, Java, HTML, CSS
Lab Equipment: DC Power Supply, Oscilloscope, Digital Multimeter,
Software: Microsoft PowerPoint, Microsoft Excel, Blender, Adobe Premiere Pro